

Student verification (CLI) for Bonus-A

1) Prove EC2 is private (no public IP)

```
aws ec2 describe-instances --instance-ids i-0b9ced7b03a5d062e --query  
"Reservations[].Instances[].PublicIpAddress"
```

```
PS C:\Users\mmauv> aws ec2 describe-instances --instance-ids i-0b9ced7b03a5d062e --query "Reservations[].Instances[].PublicIpAddress"  
[]
```

2) Prove VPC endpoints exist

```
aws ec2 describe-vpc-endpoints --filters "Name=vpc-id,Values=vpc-02659974b63ff374c"  
--query "VpcEndpoints[].ServiceName"
```

- NAT Gateway is present.

```
PS C:\Users\mmauv> aws ec2 describe-nat-gateways  
{  
  "NatGateways": [  
    {  
      "CreateTime": "2026-02-09T00:25:52+00:00",  
      "NatGatewayAddresses": [  
        {  
          "AllocationId": "eipalloc-0f8bc68040ee9b0ee",  
          "PublicIp": "52.203.6.72",  
          "AssociationId": "eipassoc-0029c2cf862f94f59",  
          "Status": "succeeded",  
          "AvailabilityZone": "us-east-1a",  
          "AvailabilityZoneId": "use1-az6"  
        }  
      ],  
      "NatGatewayId": "nat-11b81a7687a7657ef",  
      "State": "available",  
      "VpcId": "vpc-02659974b63ff374c",  
      "Tags": [  
        {  
          "Key": "Owner",  
          "Value": "Mike"  
        },  
        {  
          "Key": "ManagedBy",  
          "Value": "Terraform"  
        },  
        {  
          "Key": "Name",  
          "Value": "lab-1-regional-nat"  
        }  
      ],  
      "ConnectivityType": "public",  
      "AvailabilityMode": "regional",  
      "AutoScalingIps": "enabled",  
      "AutoProvisionZones": "enabled",  
      "RouteTableId": "rtb-02083265bdf3ea096"  
    }  
  ]  
}
```

3) Prove Session Manager path works (no SSH)

```
aws ssm describe-instance-information --query "InstanceInformationList[].InstanceId"
```

Expected: your private EC2 instance ID appears

```
ubuntu@ip-10-88-11-107:~$ ssh -i "armageddon-keys.pem" ec2-user@10.88.11.212
Warning: Identity file armageddon-keys.pem not accessible: No such file or directory.
The authenticity of host '10.88.11.212 (10.88.11.212)' can't be established.
ED25519 key fingerprint is SHA256:Drh7rM63j5uoA5n04U+vIdcjADoXYqfoox79ITYHRcQ.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.88.11.212' (ED25519) to the list of known hosts.

A newer release of "Amazon Linux" is available.
Version 2023.10.20260202:
Run "/usr/bin/dnf check-release-update" for full release and version update info

      #_
     / \_ ##### Amazon Linux 2023
    /   \_#####\
   /       \####|
  /         \#/___#
 /           \|___#> https://aws.amazon.com/linux/amazon-linux-2023
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[ec2-user@ip-10-88-11-212 ~]$ aws ssm describe-instance-information --query "InstanceInformationList[].InstanceId"
[]
```

Explanation: Not using SSM to manage the instance. I SSH via bastion instead.

4) Prove the instance can read both config stores

Run from SSM session:

```
aws ssm get-parameter --name /lab/db/endpoint
```

```
[ec2-user@ip-10-88-11-212 ~]$ aws ssm get-parameter --name /lab/db/endpoint
{
  "Parameter": {
    "Name": "/lab/db/endpoint",
    "Type": "String",
    "Value": "lab-1-rds.cwuz3uwolwom.us-east-1.rds.amazonaws.com",
    "Version": 1,
    "LastModifiedDate": "2026-02-09T00:30:51.202000+00:00",
    "ARN": "arn:aws:ssm:us-east-1:068259748700:parameter/lab/db/endpoint",
    "DataType": "text"
  }
}
```

```
aws secretsmanager get-secret-value --secret-id lab-1/rds/mysql
```

```
[ec2-user@ip-10-88-11-212 ~]$ aws secretsmanager get-secret-value --secret-id lab-1/rds/mysql
{
  "ARN": "arn:aws:secretsmanager:us-east-1:068259748700:secret:lab-1/rds/mysql-eN94VF",
  "Name": "lab-1/rds/mysql",
  "VersionId": "terraform-20260209003053508000000000",
  "SecretString": "{\"dbname\":\"rds\",\"engine\":\"mysql\",\"host\":\"lab-1-rds.cwuz3uwlom.us-east-1.rds.amazonaws.com\",\"password\":\"TK0rpxkGjy36956kJ\",\"port\":3306,\"username\":\"admin\"}",
  "VersionStages": [
    "AMSCURRENT"
  ],
  "CreateDate": "2026-02-09T00:30:51.284000+00:00"
}
[ec2-user@ip-10-88-11-212 ~]$
```

5) Prove CloudWatch logs delivery path is available via endpoint

```
aws logs describe-log-streams --log-group-name /aws/ec2/lab-1-rds-app
```

```
[ec2-user@ip-10-88-11-212 ~]$ aws logs describe-log-streams --log-group-name /aws/ec2/lab-1-rds-app
{
  "logStreams": [
    {
      "logStreamName": "i-04494f5d5464e674c-cloud-init-output",
      "creationTime": 1770597120536,
      "firstEventTimestamp": 1770597115449,
      "lastEventTimestamp": 1770597120744,
      "lastIngestionTime": 1770597125747,
      "uploadSequenceToken": "49039859661235712266684585782078461470823567775260114756",
      "arn": "arn:aws:logs:us-east-1:068259748700:log-group:/aws/ec2/lab-1-rds-app:log-stream:i-04494f5d5464e674c-cloud-init-output",
      "storedBytes": 0
    },
    {
      "logStreamName": "i-04494f5d5464e674c/rdsapp-err",
      "creationTime": 1770597120517,
      "firstEventTimestamp": 1770597115452,
      "lastEventTimestamp": 1770597120688,
      "lastIngestionTime": 1770597125691,
      "uploadSequenceToken": "49039859661235712192247818018123173763902076722393022303",
      "arn": "arn:aws:logs:us-east-1:068259748700:log-group:/aws/ec2/lab-1-rds-app:log-stream:i-04494f5d5464e674c/rdsapp-err",
      "storedBytes": 0
    },
    {
      "logStreamName": "i-0675e365c59dde70e-cloud-init-output",
      "creationTime": 1770603939500,
      "firstEventTimestamp": 1770603934310,
      "lastEventTimestamp": 1770603939721,
      "lastIngestionTime": 1770603944724,
      "uploadSequenceToken": "49039859661244776241815599220362729347116879652729028571",
      "arn": "arn:aws:logs:us-east-1:068259748700:log-group:/aws/ec2/lab-1-rds-app:log-stream:i-0675e365c59dde70e-cloud-init-output",
      "storedBytes": 0
    },
    {
      "logStreamName": "i-0675e365c59dde70e/rdsapp-err",
      "creationTime": 1770603939382,
      "firstEventTimestamp": 1770603934312,
      "lastEventTimestamp": 1770603934312,
      "lastIngestionTime": 1770603939675,
      "uploadSequenceToken": "49039859661244769530543448502322487794457428780068536264",
      "arn": "arn:aws:logs:us-east-1:068259748700:log-group:/aws/ec2/lab-1-rds-app:log-stream:i-0675e365c59dde70e/rdsapp-err",
      "storedBytes": 0
    }
  ]
}
```